# Xanthovar Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-320/C

### Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Deviations observed during the campaign were investigated and closed with no impact to product quality. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

### Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. No changes to the approved analytical procedures are proposed in this submission. Deviations observed during the campaign were investigated and closed with no impact to product quality. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

The control strategy links critical quality attributes to critical process parameters identified during development. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Cleaning verification swab results were below the health-based exposure limits derived for the product.

## Control of Excipients (3.2.P.4)

For the commercial formulation F-320/C, Avicel PH-102 (Microcrystalline cellulose, filler-binder) is used.

## Grade Selection Rationale

The basis for selection of Avicel PH-102 is reduced lot-to-lot variability reported by the qualified supplier, demonstrated across three pilot-scale batches.

| Attribute | Acceptance approach |
| --- | --- |
| Identity | Compendial monograph |
| Assay / purity | Compendial monograph |
| Functionality-related characteristics | Supplier certificate plus in-house verification* |

* Control approach per IPEC Excipient Qualification Guide, ICH Q8(R2), Ph. Eur. general monograph 2034.

## Registered Specification

| Component | Function | Grade per current specification |
| --- | --- | --- |
| Microcrystalline cellulose | filler-binder | Avicel PH-200 |
| Reference | | SPEC-EX-5280 Rev 06 |

## Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Deviations observed during the campaign were investigated and closed with no impact to product quality. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

## Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Deviations observed during the campaign were investigated and closed with no impact to product quality. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Cleaning verification swab results were below the health-based exposure limits derived for the product. The control strategy links critical quality attributes to critical process parameters identified during development.